EviSearch: A Human in the Loop System for Extracting and Auditing Clinical Evidence for Systematic Reviews

[1]Naman Ahuja [1]Saniya Mulla [2]Muhammad Ali Khan [2]Zaryab Bin Riaz
[2]Kaneez Zahra Rubab Khakwani [2]Mohamad Bassam Sonbol [2]Irbaz Bin Riaz [1]Vivek Gupta
Arizona State University[1] Mayo Clinic[2]
riaz.irbaz@mayo.edu, vgupt140@asu.edu

Abstract

We present **EviSearch**, a multi-agent extraction system that automates the creation of ontology-aligned clinical evidence tables directly from native trial PDFs while guaranteeing per-cell provenance for audit and human verification. EviSearch pairs a PDF-query agent (which preserves rendered layout and figures) with a retrieval-guided search agent and a reconciliation module that forces page-level verification when agents disagree. The pipeline is designed for high-precision extraction across multimodal evidence sources (text, tables, figures) and for generating reviewer-actionable provenance that clinicians can inspect and correct. On a clinician-curated benchmark of oncology trial papers, EviSearch substantially improves extraction accuracy relative to strong parsed-text baselines while providing comprehensive attribution coverage. By logging reconciler decisions and reviewer edits, the system produces structured preference and supervision signals that bootstrap iterative model improvement. EviSearch is intended to accelerate living systematic review workflows, reduce manual curation burden, and provide a safe, auditable path for integrating LLM-based extraction into evidence synthesis pipelines.

1 Introduction

Structured extraction of clinical evidence from trial publications is a foundational step in evidence synthesis, meta-analysis, and clinical guideline development. Living systematic review platforms[1] curate interactive tables summarizing trial identifiers, treatment arms, patient characteristics, endpoints, and subgroup outcomes, enabling clinicians to compare therapies at a glance. However, constructing and maintaining these structured evidence tables remains a largely manual process requiring expert review of full-text PDFs. Recent advances in large language models (LLMs) have demonstrated strong performance across medical reasoning, documentation, and summarization tasks (Zhou et al., 2023; Wang et al., 2025). Clinical evaluations and systematic reviews highlight the potential of LLMs to support diagnostic reasoning and information synthesis, while also revealing substantial variability and reliability concerns in real-world settings (Omar et al., 2024; Meng et al., 2024).

In the context of clinical trial extraction, hallucinated or misattributed values can directly impact downstream meta-analyses and treatment decisions. Beyond reliability, clinical trial PDFs present intrinsic technical challenges. Evidence is distributed across heterogeneous modalities: narrative text, complex tables, Kaplan–Meier plots, and figure captions. Some schema fields require global document-level reasoning (e.g., determining whether quality-of-life outcomes were reported), while others demand fine-grained cell-level extraction from structured tables or graphical figures. Subgroup reporting (e.g., high- vs. low-volume disease) introduces scope and normalization challenges that exceed simple pattern matching.

We present **EviSearch**, a multi-agent extraction framework designed to automate ontology-aligned evidence tables directly from native clinical trial PDFs while enforcing per-cell provenance and verification. EviSearch combines a direct PDF query agent that preserves multi-modal layout with a retrieval-guided search agent targeting page-level evidence. A reconciliation module adjudicates disagreements via forced page-level verification, and a human-on-the-loop interface exposes grounded attribution for auditing and feedback. We evaluate EviSearch on a clinician-curated benchmark whose schema mirrors the fields used in living evidence platforms for metastatic castration-sensitive prostate cancer (mCSPC). Our results show substantial improvements over baselines while achieving comprehensive attribution coverage.

We encourage readers to try it at the following

[1]https://mcspc.living-evidence.com/

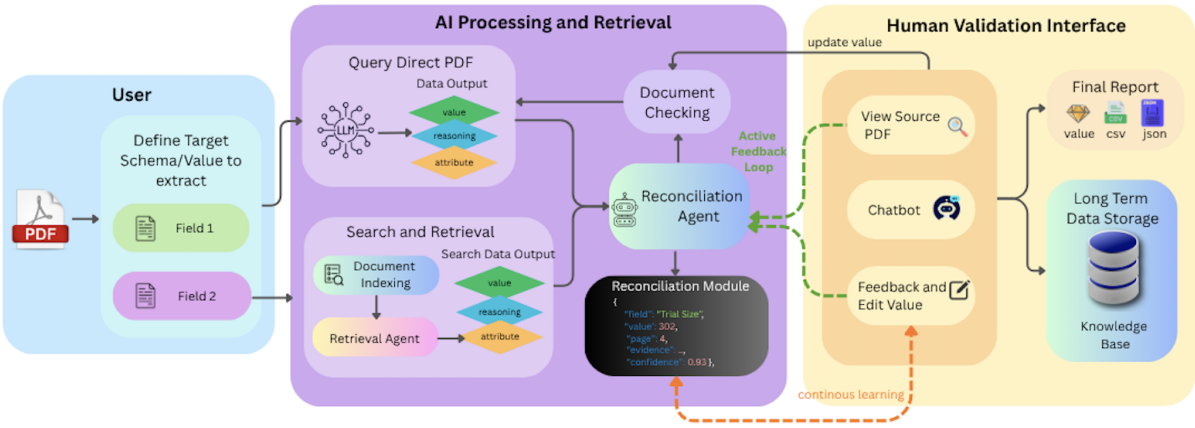

Figure 1: EviSearch system architecture

link:

- Demo: https://coral-lab-asu.github.io/EviSearch/
- Video

2 Related Work

Understanding scientific documents has been widely studied in natural language processing and document AI. Domain-adapted pretraining models such as SciBERT improve representation learning for scholarly text (Beltagy et al., 2019), while large-scale corpora like S2ORC enable structured modeling of scientific papers at scale (Lo et al., 2020). Document question answering benchmarks such as DocVQA evaluate models on visually rich PDFs requiring joint reasoning over text and layout (Mathew et al., 2021), and layout-aware architectures including LayoutLM encode spatial structure alongside textual content to improve document understanding (Xu et al., 2020). OCR-free transformers such as Donut further advance end-to-end document reasoning without explicit text extraction (Kim et al., 2022). In parallel, structured table reasoning has progressed through models such as TAPAS, TaBERT, and TURL, which incorporate tabular representations into transformer architectures for semantic parsing and question answering (Herzig et al., 2020; Yin et al., 2020; Deng et al., 2020). Chart and plot understanding benchmarks, including ChartQA and PlotQA, demonstrate the visual and logical reasoning required to interpret graphical data (Masry et al., 2022; Methani et al., 2020). Retrieval-augmented generation further grounds model outputs in external evidence to improve factual consistency (Lewis et al., 2020), and recent multimodal LLMs extend this paradigm to documents and images. However, existing approaches typically address isolated components of the problem, document layout modeling, table reasoning, chart interpretation, or retrieval-based grounding, rather than providing a unified system that performs schema-constrained extraction across multimodal scientific PDFs with explicit per-cell provenance and auditable verification. EviSearch brings these strands together into a single pipeline designed specifically for ontology-aligned clinical evidence synthesis.

3 EviSearch System Architecture

EviSearch is a multi-stage, multi-agent extraction pipeline that fills a structured clinical evidence table from a trial publication PDF, while producing grounded, auditable attribution for every extracted value. The pipeline consists of four stages: (i) document parsing and chunking, (ii) parallel extraction by two independent agents, (iii) automated reconciliation, and (iv) human review and feedback through a web interface. Figure 1 provides an overview of the full architecture.

3.1 Column Schema and Batching

The output schema consists of 133 columns drawn from the structured evidence tables used in the LISR living evidence synthesis platform [2]. Each column is paired with a natural-language definition that specifies the required value, reporting conventions, and fallback behavior (e.g., "use *Not reported* if missing"). Columns are annotated with an evaluation category (`numerical`, `free-text`) and grouped by clinical section: trial characteristics,

[2] https://mcspc.living-evidence.com/

population characteristics, efficacy outcomes, subgroup reporting, adverse events, and demographic breakdowns.

To enable parallelism while maintaining inter-column context, columns are batched using a group-aware packing algorithm: columns within the same clinical section are kept together, and batches are limited to a maximum of 15 columns to balance context length against reliability. Groups exceeding this limit are split into sequential sub-batches; smaller groups are merged greedily until the batch limit is reached. Both extraction agents and the reconciliation agent operate over the same batch structure, enabling direct comparison at the column level.

3.2 Document Parsing and Retrieval Indexing

Each PDF is parsed using Landing AI's `dpt-2-latest` Document Parse model, which produces element-level chunks with associated page numbers and modality labels (`text`, `table`, `figure`). The output is stored as structured JSON and a rendered Markdown representation preserving table layout for downstream use.

For semantic retrieval, each chunk is embedded using OpenAI's `text-embedding-3-large` model (3,072 dimensions), producing a per-document index over page-level chunks. Embeddings are computed in batches of 100 and indexed for cosine similarity search. This index is used exclusively by the Search Agent (§3.4) and is not shared with the PDF Query Module.

3.3 PDF Query Module (Agent A)

The PDF Query Module submits the full PDF binary alongside each column batch to Gemini-2.5-Flash via the File API. This preserves the document's native structure including figures, multi-column layouts, and table formatting, without relying on parsed text. The model is prompted with the column definitions and instructed to return, for each column, a structured tuple of `(value, reasoning, attribution: {page, modality, verbatim_quote})`. Outputs are constrained via tool-based structured generation (function calling with a strict JSON schema), and temperature is set to 0 for deterministic extraction.

This module is most effective on columns requiring global document context, such as trial design, eligibility criteria, and qualitative judgments Because the PDF is re-uploaded per API call, input

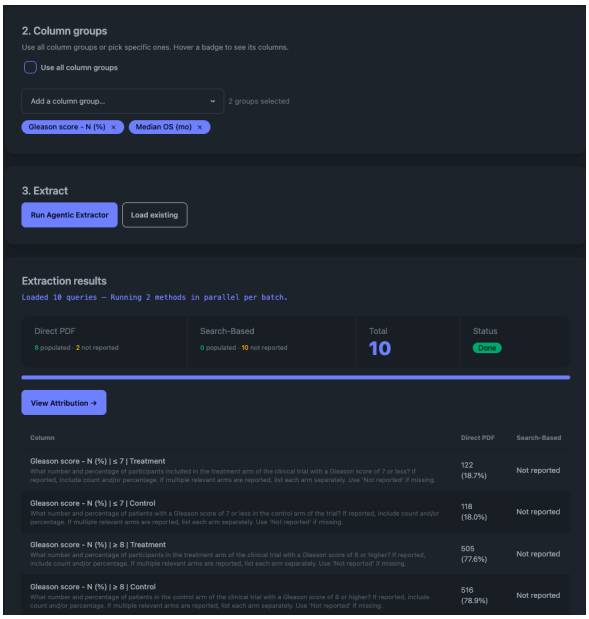

Figure 2: Extraction Interface

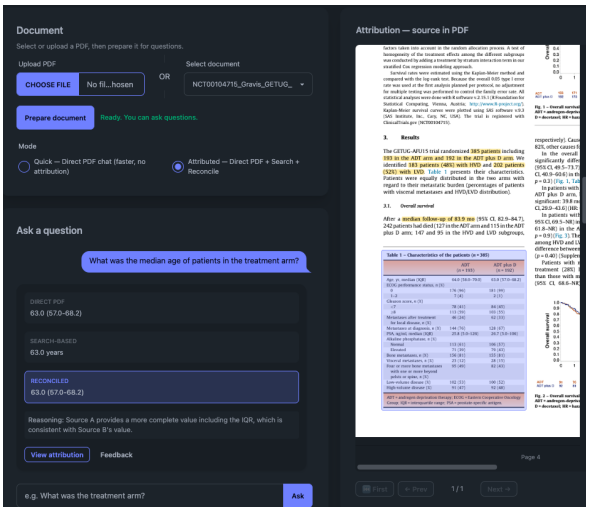

Figure 3: Attribution Interface

token counts include document tokens at each invocation; this is reflected in the cost analysis (§5.3).

3.4 Search Agent (Agent B)

The Search Agent operates over the same column batches using a tool-based agentic loop over the parsed document representation, targeting fine-grained evidence in tables, figures, and results sections. It is powered by Gemini-2.5-Flash at temperature 0, with access to three tools:

- `search_chunks`: Performs semantic search over the document index using the column definition as the query, returning the top-k ($k = 5$) most relevant pages with full content.

- `get_chunks_by_page`: Loads the full parsed content of specified pages by number, enabling targeted follow-up.

- `submit_extraction`: Submits the final extracted values with structured attribution once all columns are resolved.

The agent follows an extract-first policy: it attempts extraction from the initial batch context before invoking retrieval tools for unresolved columns. A global deduplication mechanism tracks pages already provided to the agent within a session; subsequent requests for the same page return a cache pointer rather than re-sending content, bounding the effective context size and preventing redundant token expenditure. Outputs include a `(value, reasoning, attribution)` tuple for each column, with attribution normalised to `{page, modality}` where modality $\in$ `{text, table, figure}`.

3.5 Reconciliation Agent

The Reconciliation Agent receives the outputs of Agent A and Agent B for each column batch and adjudicates the final value. It applies a two-pass protocol:

Pass 1: Agreement detection (no tool use): Columns are resolved immediately if: (a) both agents report *Not reported*; (b) both values are identical; or (c) both agents agree on page and modality and one value is a strict superset of the other (e.g., a more complete numeric expression). These are assigned `both_correct`.

Pass 2: Verified resolution (forced tool use): In case of conflicts, where one agent reports a value and the other reports *Not reported*, or where values differ: require the agent to call `get_page` before submitting. This tool returns both the full parsed text and a rendered page image for the disputed page, enabling multimodal verification. The agent then submits a reconciled value with one of four verification labels: `both_correct`, `A_correct_B_wrong`, `B_correct_A_wrong`, or `both_wrong`. Outputs are validated against a strict schema before being written to disk.

Columns labelled `both_wrong` indicate that neither agent produced a verifiable answer; these are surfaced as low-confidence in the review interface for targeted human attention.

3.6 Human Review, Auditability, and Verification Modes

EviSearch is designed around a *human-on-the-loop* principle: the system operates fully autonomously by default, but every extraction decision is grounded in verifiable evidence that a human expert can inspect, challenge, or correct at any level of granularity. This is achieved through two complementary verification workflows.

Automated mode. For the common case, the reconciliation agent resolves all columns independently. Each extracted value is stored with explicit provenance; source page, modality, verbatim quote, and the reconciler's reasoning and rendered in the web interface with chunk-level highlighting directly in the source PDF. Clinicians receive a completed table in which every cell is one click away from its evidence.

Human-assisted mode. For columns the system flags as uncertain (`both_wrong`), or any column a reviewer chooses to inspect, the same evidence infrastructure used by the reconciliation agent is available to the human directly. The reviewer sees Agent A's answer, Agent B's answer, the reconciler's judgment, and the attributed page content side by side, and can accept one candidate or write a corrected value. Critically, the human is never auditing a black box: they review the same grounded evidence chain the reconciler produced.

4 Experiments

Dataset: We evaluate EviSearch on a benchmark of peer-reviewed clinical trial papers sampled from our clinician-curated dataset from the ontology of metastatic castration-sensitive prostate cancer (mCSPC). For mCSPC, the clinical ontology encompasses standardized trial characteristics including trial identifiers, treatment and control arms, systemic therapy regimens, comparator groups, patient population summaries (e.g., age, performance status, disease volume), outcome measures (e.g., overall survival, progression-free survival), and subgroup results stratified by prognostic features such as high- vs. low-volume disease, synchronous vs. metachronous metastases. This mirrors the structured evidence presented in living systematic reviews of first-line mCSPC therapies, which summarize study design, patient characteristics, and comparative outcomes in tabular structures. Each paper in the benchmark is annotated with a structured schema of columns covering these clinical

Method	Numeric Columns			Free-Text Columns			Overall (All)		
	Corr.	Comp.	Ovrl.	Corr.	Comp.	Ovrl.	Corr.	Comp.	Ovrl.
Gemini 2.5 Flash (PDF upload)	83.5	81.7	82.6	78.3	76.7	77.5	82.5	80.9	81.7
Gemini 2.5 Flash (parsed Doc)	79.4	77.4	78.4	72.0	71.7	71.8	78.3	76.7	77.5
GPT-4.1 (parsed Doc)	84.7	86.5	85.6	76.7	78.0	77.3	83.3	84.9	84.1
EviSearch (Ours)	**91.0**	**92.3**	**91.7**	**89.7**	**87.7**	**88.7**	**90.9**	**91.6**	**91.3**

Table 1: Extraction performance comparison across methods. **Corr.** = Correctness, **Comp.** = Completeness, **Ovrl.** = Overall (mean of Corr. and Comp.). All scores are in %.

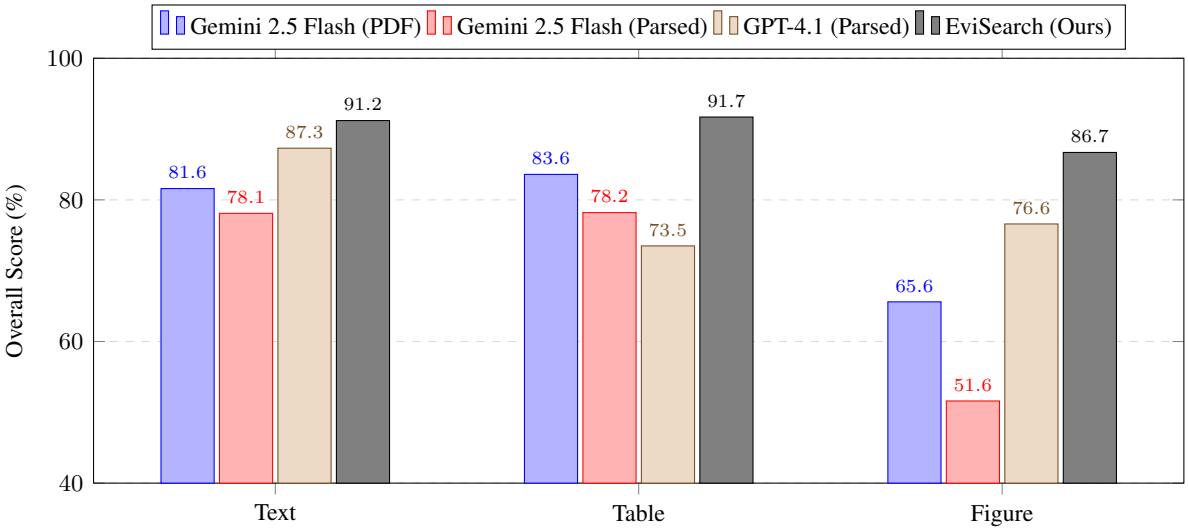

Figure 4: Overall extraction performance across evidence modalities.

elements, and for every reported value, clinicians provide gold-standard evidence attribution including source page and modality (text, table, figure).

4.1 Baselines:

We compare EviSearch against three baselines. (1) *Gemini 2.5 Flash (PDF upload)*, which uploads pdf and extracts structured outputs (2) *Gemini 2.5 Flash (parsed Doc)*, which extracts values from Landing-AI-generated markdown text extracted from the PDF during pre-processing. (3) *GPT-4.1 (parsed Doc)*, applied to the same parsed markdown representation.

Evaluation For evaluation, all columns are divided into two categories, numerical and free text. We use an LLM judge with specifically crafted instructions for both categories to evaluate predictions against ground-truth outputs. We report three metrics: *Correctness*, the proportion of extracted values matching the gold standard; *Completeness*, the proportion of required schema columns successfully filled; and *Overall*, defined as the mean of correctness and completeness. For numeric fields, we apply tolerant matching to account for rounding and formatting differences.

5 Results and Analysis

5.1 Overall Extraction Performance

Table 1 reports extraction performance across all four methods on our benchmark. EviSearch achieves the highest scores across all metrics and both column types: 91.3% overall (90.9% correctness, 91.6% completeness), compared to the best baseline GPT-4.1 (parsed Doc) at 84.1%, a 7.2 point gain. This gap is meaningful in a clinical context, where missed or incorrect values can propagate into downstream meta-analyses.

Among the baselines, GPT-4.1 attains 85.6% on numeric columns but drops to 77.3% on free-text, a 8.3 point gap suggesting that it relies more on surface-level pattern matching than on genuine semantic understanding of clinical language. Gemini 2.5 Flash (parsed Doc) performs worst overall (77.5%), confirming that feeding a model parsed markdown, even with well-preserved structure, is insufficient when evidence is spread across heterogeneous modalities. Notably, Gemini 2.5 Flash improves by 4.2 points when given the native PDF

Method	In Tok.	Out Tok.	Total Tok.	API Calls
Gemini (Native PDF)	135,564	13,930	149,494	39
EviSearch	603,350	39,448	642,798	79
Gemini (Parsed Text)	957,761	12,521	970,282	39
GPT-4.1 (Parsed Text)	990,819	11,107	1,001,926	39

Table 2: Average token usage and API calls per document.

directly (81.7%), highlighting the value of preserving document layout and figures.

5.2 Performance by Evidence Source Modality

Of the 667 evidence fields in our benchmark, 53.4% originate from text passages, 41.8% from tables, and 4.8% from figures, meaning over 46% of the evidence cannot be recovered from plain text alone. This distribution reflects the real structure of clinical trial papers, where key outcomes are often reported in dense result tables or survival curves rather than narrative prose. Figure 4 breaks down performance by source modality as determined by gold-standard attribution.

The modality breakdown reveals where each system structurally struggles. All baselines degrade on figure-sourced evidence: Gemini 2.5 Flash (parsed Doc) drops sharply to 51.6%, a gap of 26.5 points behind EviSearch (86.7%) on the same category. This confirms that vanilla text-based prompting simply cannot recover values embedded in charts or plots. Table extraction further separates the methods, GPT-4.1 drops 13.8 points moving from text to table fields (87.3 $\rightarrow$ 73.5), likely because clinical result tables involve complex multi-row, multi-column structures that require targeted retrieval rather than global document context. EviSearch, by contrast, maintains near-constant performance across all three modalities (91.2 $\rightarrow$ 91.7 $\rightarrow$ 86.7), a robustness that directly reflects the complementary strengths of its two agents: the PDF query module handles layout-sensitive and figure-rich content, while the search agent specialises in structured table retrieval.

5.3 API Cost and Efficiency

Table 2 shows average token usage and API calls per document. EviSearch uses 642,798 total tokens over 79 API calls, compared to 149,494 for Gemini native and $\sim$1M tokens for the parsed baselines. The higher cost reflects the dual-agent and reconciliation architecture, as each column batch is processed independently by two extraction agents and subsequently verified through forced page-level adjudication when disagreements arise. In addition, per-cell provenance storage and structured tool-based generation introduce modest overhead relative to single-pass prompting.

However, this additional computation is directly tied to auditability guarantees. Unlike parsed-text baselines that consume large token budgets without preserving grounded attribution, EviSearch converts its token expenditure into verifiable evidence chains: every extracted value is traceable to a specific page and modality, and disagreements trigger explicit multimodal verification rather than silent failure. Notably, EviSearch remains substantially more efficient than parsed-text pipelines, which exceed 970k–1M tokens per document due to repeated inclusion of full document context. The retrieval-guided Search Agent and deduplicated page caching constrain context growth, ensuring that additional cost scales with uncertainty rather than document length.

From a deployment perspective, the cost profile reflects a deliberate tradeoff: modestly higher API usage than native PDF prompting in exchange for significant gains in accuracy (+7.2 overall) and near-complete attribution coverage. In clinical evidence synthesis workflows, where extraction errors can propagate into downstream meta-analyses, this cost–accuracy tradeoff favors reliability and auditability over minimal token usage.

6 Conclusion

We introduced EviSearch, a human-on-the-loop system combining direct PDF querying, retrieval-guided extraction, and principled reconciliation to produce accurate, auditable clinical evidence tables from trial publications. EviSearch substantially outperforms strong baselines while providing near-complete provenance coverage, with every extracted value traceable to its source.

Beyond accuracy, EviSearch acts as a data flywheel: reviewer corrections and reconciliation decisions generate structured supervision signals for continual improvement. We envision it as a practical step toward trustworthy automation in living systematic reviews, reducing manual burden while preserving clinical oversight and reproducibility.

Limitations

While EviSearch demonstrates strong extraction performance, several limitations remain. The system relies on LLMs whose outputs are probabilistic and may exhibit reasoning errors or misinterpretation of nuanced clinical terminology. Evaluation uses an LLM-based judge, which improves scalability but may not capture all clinically meaningful distinctions relative to full expert adjudication. Token costs are non-trivial due to the dual-agent architecture, potentially limiting large-scale deployment. Finally, full automation of clinical evidence synthesis is neither the goal nor currently advisable; responsible use requires expert oversight, particularly when outputs inform meta-analyses or clinical decisions.

Ethics Statement

EviSearch processes publicly available, peer-reviewed trial PDFs that contain no personally identifiable patient information. The system stores explicit provenance for every extracted value to enable traceability and error correction; reviewer edits are logged as structured feedback but no private user data is retained. EviSearch relies on third-party LLM APIs (e.g., Gemini), and users must supply their own credentials. Software components are released under the Apache 2.0 License; responsible deployment requires adherence to applicable data governance policies and clinical oversight standards.

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